Video Transcript

Video 8: Reinforcement Learning Solution (02:39)

So, the goal was to train the robot so that it could find the goal, which is room zero here. And, as a sample, we'll start it at room eight. So, the first thing we need to do is to get the geometry into the reward matrix. So, this is the reward matrix that I found. So, on row zero is connected to one and to three, and one is connected to zero, two, and four, etc. So, we need to get the geometry correct and once we've got that, we can put it into our code. I'll show you the code that I have.

We start out with NumPy, and we need to get our location sorted out. This is just syntactic sugar really. So, here's the new room layout. And now the code, I'm running with gamma 0.75 and alpha 0.9. This code stays exactly the same. I put 100 in there for the reward at the ending state, the goal, but everything else is the same. And, the only thing we need to alter is the start and the end point. And, I'm starting at eight and going to room zero. And, here's my Q-matrix being printed out. I'll run this just in case you ran it with one and 0.5 for these. So, gamma 0.5 and alpha as 1.0. Let's just run it and we should get a very quick solution here. So, this is with gamma as 0.5 and alpha as 1.0. But we get basically the same root, so the robot does just as well with either of those choices. Okay, I hope you enjoyed learning about reinforcement learning; it's a very powerful technique indeed. Okay, bye for now.

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